

POKHARA UNIVERSITY

Level: Bachelor
Semester: Fall
Programme: BE
Course: Numerical Methods

Year : 2022
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Find the root of the equation $f(x) = x^2 - 4x - 10$ correct to three decimal places by using False Position method. 7
- b) Estimate the root of the equation $f(x) = xe^x - \cos x$ using Newton Raphson method correct to three decimal places. 8
2. a) From the following table estimate the number of student who obtained marks between 40 and 45. 7

Marks	30-40	40-50	50-60	60-70
No of Students	31	42	51	35

- b) From the following data given in the table below evaluate $f'(2.5)$ by using Lagrange method. 8

x	1	2	4	5	7
f(x)	1	1.414	1.732	2.00	2.6

3. a) Evaluate $\int_1^5 \frac{1}{x} dx$ by using Gaussian Integration formula for $n=3$ and compare the value with exact solution. 7
- b) Use the Romberg integration to find the solution correct upto three decimal places. 8

$$I = \int_0^1 \frac{1}{1+x^2} dx$$

4. a) Find the solution of the given simultaneous linear equation using Gauss Seidel method. 7
- $$\begin{aligned} 6x_1 - 2x_2 + x_3 &= 11 \\ -2x_1 + 7x_2 + 2x_3 &= 5 \\ x_1 + 2x_2 - 5x_3 &= 1 \end{aligned}$$

- b) Solve the following system of equations using Crout method. 8
- $$x + y + z = 4, x + 4y + 3z = 8, x + 6y + 2z = 6$$

5. a) Using the Euler's (R-K 1st order method) find an approximate value of y corresponding to $x=1$, given that $dy/dx = X+Y$ and $y=1$. When $x=0, h=0.1$. 8

- b) Apply Euler's method to approximate value of $y(0.3)$ for the differential equation: 7

$$\frac{dy}{dx} = y + x, y(0) = 1.$$

6. a) Torsion on a square bar of size 15cm* 15cm. If two of the sides are held at 100°C and the other two sides are held at 0°C. Calculate the steady state temperature at interior points. Assume a grid size of 5cm * 5cm. 8

- b) Solve the Poisson equation $\nabla^2 f = 2x^2 + y$, over the square domain $1 \leq x \leq 4, 1 \leq y \leq 4$, with $f=0$ on the boundary. Take step size in x and $y, h=k=1$. 7

7. Write short notes on: (Any two) 2*5

- a) Ill-conditioned and Well-conditioned systems
- b) Error in Numerical method
- c) Cubic Spline.

POKHARA UNIVERSITY

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Attempt all the questions.

Explain in brief the errors in numerical calculations. 5

Find a root of $3x + \sin x - e^x = 0$ using 10

- One of the bracketing methods and
- One of the non-bracketing methods

From the data given below, find the number of students whose weight is between 60 to 70. 8

Weight in lbs	0-40	40-60	60-80	80-100	100-120
No. of students	250	120	100	70	50

Using the method of least squares, fit the curves $ax^2 + \frac{b}{x}$ to the following data. 7

x	1	2	3	4
y	-1.52	0.96	8.88	7.66

Use Romberg's method, to compute $\int_0^2 \frac{e^x + \sin x}{1+x^2} dx$ correct up to two decimal places. 8

Estimate approximate derivative of $f(x) = x^2$ at $x=1$ for $h=0.1, 0.2, 0.05, 0.01$. Use first order difference method and find the respective error. 7

4. a) Apply the factorization method to solve the equation $3x+2y+7z=4$; $2x+3y+z=5$; $3x+4y+z=7$ 8

b) Using SOR method, solve the following system of $4x+y+2z=4$; $3x+5y+z=7$; $x+y+3z=3$. 7

5. a) Find the largest eigen value and the corresponding eigen vector of the matrix $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ using power method 8

b) Using the R-K 1st order method, find an approximate value of y corresponding to $x=1$, given that $\frac{dy}{dx} = \frac{y-x}{y+x}$ and $y=1$. When $x=0$, and $h=0.02$. 7

6. a) Using the R-K method of fourth order, solve for y at $x=1.2, 1.4$, from $\frac{dy}{dx} = \frac{2xy+e^x}{x^2+xe^x}$ given $x_0=1, y_0=0$. 8

b) Solve the elliptic equation $U_{xx} + U_{yy} = 0$ over a square mesh of side four units satisfying the following boundary conditions; $u(0,y)=0$ for $0 \leq y \leq 4$; $u(4,y)=12+y$ for $0 \leq y \leq 4$; $u(x,0)=3x$ for $0 \leq x \leq 4$; $u(x,4)=x^2$ for $0 \leq x \leq 4$ 7

7. Write short notes on: (Any two) 2x5

- Shooting Method
- Algorithm of Gauss Jordan method
- Algorithm of fixed point iteration method

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Numerical Methods (Old)

Semester: Fall

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Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) Find where the graph of $y = x-3$ and $y = \ln(x)$ intersect using bisection method. Get the intersection value correct to four decimal places. 8

- b) Find value of $\sqrt{18}$ using Newton Raphson method. 7

2. a) Find the root of the equation $3x + \sin x - e^x = 0$ corrected up to 3 decimal places using Secant Method. 7

- b) Given populations of a city in different year as shown in table below. 8

x (in year)	1891	1901	1911	1921	1931
f(x)(population of that year in 0000)	46	66	81	93	101

Observe the data carefully and answer the following questions:

- Which method is applicable for finding population in the year 1860 and why? Also find the population of that year.
- If you are asked to find the change in population between the year 1925 to 1930, how will you find? (mention formula and steps)

3. a) Fit a power function model of the form $y = ax^b$ from the given data points. 8

x	1	2	3	4	5
y	0.5	2	4.5	8	12.5

- b) Use Romberg method to evaluate: 7

$$\int_0^{3\pi/2} e^x \sin x \, dx$$

4. a) Solve the following system of equations using partial pivoting method. 8
- $$x + y + z = 4, \quad x + 4y + 3z = 8, \quad x + 6y + 2z = 6$$

- b) Obtain the solution of following system using Gauss-Seidel iteration method. 7

$$2x_1 + x_2 + x_3 = 5$$

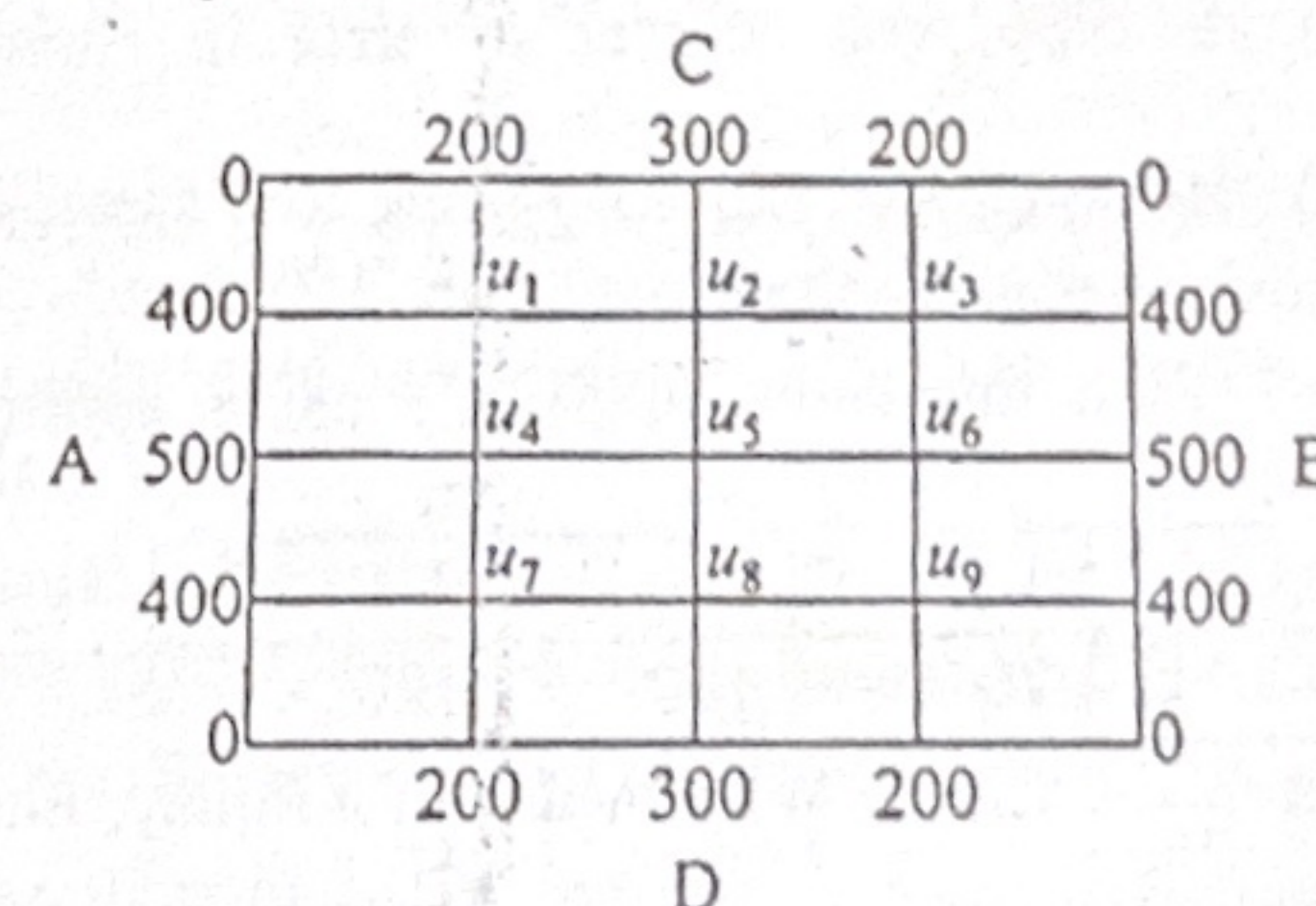
$$3x_1 + 5x_2 + 2x_3 = 15$$

$$2x_1 + x_2 + 4x_3 = 8$$

5. a) Determine the largest eigenvalue and the corresponding eigenvector of the matrix: 7

$$A = \begin{bmatrix} 1 & -3 & 2 \\ 4 & 4 & -1 \\ 6 & 3 & 5 \end{bmatrix} \text{ using the power method.}$$

- b) Solve the Laplace equation $U_{xx} + U_{yy} = 0$ for the square mesh with boundary values as shown in the figure. 8



6. a) Solve the following equation for $y(0.2)$ using shooting method: 9

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 3y = 6x \text{ Given } y(0) = 0, y'(0) = 1.$$

- b) Use Picard's method to approximate the value of y when $x=0.1, x=0.2$ and $x=0.4$, given that $y=1$ at $x=0$ and $dy/dx = 1+xy$ correct to three decimal places (Use upto second approximations). 6

7. Write short notes on: (Any two) 2x5

- Newton's backward differentiation formulas
- Partial differential equation and their examples
- Algorithm of fixed point iteration method

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Numerical Methods (New)

Semester: Fall

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Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) Evaluate a real root of the given equation: $f(x) = 3x + \sin x - 2$ using Fixed point iterative method correct to four decimal places. 7

OR

Define error. Explain its type with sources of error in Numerical computation.

- b) Calculate the root of the equation: $4x^3 - 2x - 6 = 0$ correct upto three decimal places using Bisection Method. 8
2. a) What is interpolation? Find the value of $f(1.2)$ using appropriate interpolation technique. 8

x	1	1.4	1.8	2.2
f(x)	50	70	100	120

- b) If P is pull required to lift a load W by means of a Pulley, Find the law of form $P = mW + C$, (where m and C are constants) using least square method for the following data: 7

P	12	15	21	25
W	50	70	100	120

OR

Use the suitable method to fit a curve $y = ax^b$ for the following data

x	-2	-1	0	1	2	3	4
y	38	6	0	-5	-41	130	300

3. a) Compute the following using Simpson's 1/3 rule for $n=8$ with an accuracy to five decimal places. 7

$$\int_1^5 e^{-x^2} dx$$

- b) Evaluate the following using Gaussian three point Integration formula: 8

$$\int_2^4 (x^4 + 1) dx$$

4. a) Solve the following system of Linear equations using partial pivoting method. 7

$$x + y + z = 4, x + 4y + 3z = 8, x + 6y + 2z = 6$$

- b) Solve the following system of equations using Gauss Jacobi's method. 8

$$3x + 2y + z = 10, 2x + 3y + 2z = 14, x + 2y + 3z = 14$$

5. a) Find the largest Eigen value and Corresponding Eigen vector of given matrix using power method. 7

$$A = \begin{bmatrix} 2 & 5 & 1 \\ 5 & -2 & 3 \\ 1 & 3 & 10 \end{bmatrix}$$

- b) Solve the following differential equation for $y(0.4)$ using Heun's method. 8

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 3y = 6x; \text{ with } y(0) = 0 \text{ and } y'(0) = 1 \text{ (take } h=0.2).$$

6. a) Use Euler's method to solve the following equation for $y(1)$ using $h = 0.25$. 7

$$\frac{dy}{dx} = x + y + xy; y(0)=1$$

- b) Solve the Poisson equation $\nabla^2 f = -10(x^2 + y^2 + 10)$ over the square with $0 \leq x \leq 3; 0 \leq y \leq 3$ and $f = 0$ on boundary. Use $h = 1$. 8

7. Write short notes on: (Any two) 2x5

- a) Initial Value problems and Boundary value problems
b) Algorithm for secant method
c) Schmidt method for one dimensional heat equation

POKHARA UNIVERSITY

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The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Find the root of the equation $\cos x + e^x + x^2 = 3$ correct to three decimal places by using Newton Raphson method.
- b) Find the root of the equation $f(x) = x^2 - 4x - 10$ correct to three decimal places by using False Position method.

2. a) The following table gives the displacement, x (cms.) of an object at various time, t (seconds). Find the displacement of this object at $t=1.3$ second and $t=1.5$ second, using any suitable interpolation formula.

t	1.0	1.2	1.4	1.7
x	9.0	9.5	10.2	11.0

- b) The growth of bacteria(N) in a culture after t hours is given by the following table:

Time t (hr)	0	1	2	3	4
Bacteria(N)	32	47	65	92	132

If the relationship between bacteria N and time t is of the form $N=ab^t$. Using least square approximation estimate the N at $t=5$ hr.

3. a) Find $\int_0^3 \frac{e^x}{1+x} dx$ by using:
 - i. Trapezoidal rule
 - ii. Simpson's 1/3 rule
 - iii. Simpson's 3/8 rule
- b) Use the Romberg integration to find the solution correct upto three decimal places.

$$I = \int_0^1 \frac{1}{1+x^2} dx$$

4. a) Find the inverse of the square matrix, $A = \begin{bmatrix} 4 & 1 & 2 \\ 2 & 3 & -1 \\ 1 & -2 & 2 \end{bmatrix}$ by using Gauss-Jordan elimination method.

- b) Find the solution of the given simultaneous linear equation using Gauss Seidel method.

$$6x - 2y + z = 11$$

$$-2x + 7y + 2z = 5$$

$$x + 2y - 5z = -1$$

5. a) solve the following differential equation within $0 \leq x \leq 0.5$ using Rk 4th order method $10d^2y/dx^2 + 2dy/dx - 3y = 5$, $y(0)=0$, $y'(0)=0$.

- b) Using the Euler's (R-K Ist order method) find an approximate value of y corresponding to $x=1$, given that $dy/dx = X+Y$ and $y=1$. When $x=0$, $h=0.1$.

6. a) Torsion on a square bar of size $15\text{cm} \times 15\text{cm}$. If two of the sides are held at 100°C and the other two sides are held at 0°C . Calculate the steady state temperature at interior points. Assume a grid size of $5\text{cm} \times 5\text{cm}$.

- b) Solve the Poisson equation $\nabla^2 f = 2x^2 + y$, over the square domain $1 \leq x \leq 4$, $1 \leq y \leq 4$, with $f=0$ on the boundary. Take step size in x and y , $h=k=1$.

7. Write short notes on: (Any two)

2×5

- a) Applications of Numerical Methods in Engineering
- b) Error in Numerical Method
- c) Ill condition and well-conditioned system

POKHARA UNIVERSITY

Level: Bachelor

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Programme: BE

Full Marks : 100

Course: Numerical Methods (New)

Pass Marks : 45

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Attempt all the questions.

1. a) Solve $x \log_{10} x = 1.2$ by Newton- Raphson method correct to four decimal places. 8
- b) Using Secant Method, find the roots of function $2x - \log_{10} x - 7 = 0$ Correct up to three decimal places. 7

OR

Find the root of the equation $f(x) = x^2 - 4x - 10$ correct to three decimal places by using False Position method.

2. a) From the following data given in the table below evaluate $f'(2.5)$ by using Lagrange method. 8

x	1	2	4	5	7
f(x)	1	1.414	1.732	2.00	2.6

- b) From the following table, Estimate the number of student who obtained marks between 50 and 55. 7

Marks	30-40	40-50	50-60	60-70	70-80
No of Students	31	42	51	35	31

3. a) Compute the Simpson's 1/3 and Simpson's 3/8 rule for $I = \int_0^1 e^{-x^2} dx$ using a regular partition with subinterval $n=6$. 8
- b) Use the Romberg integration to find the solution correct upto three decimal places. 7

$$I = \int_0^1 \frac{1}{1+x^2}$$

$$4. \quad 3x + 2y + z = 10$$

$$a) \quad \text{Solve : } 2x + 3y + 2z = 14$$

$$x + 2y + 3z = 14$$

By Gauss elimination method.

- b) Solve the following system of equations using Crout method. 7

$$x + y + z = 4, x + 4y + 3z = 8, x + 6y + 2z = 6$$

OR

Find largest eigen value and corresponding eigen vector of the matrix.

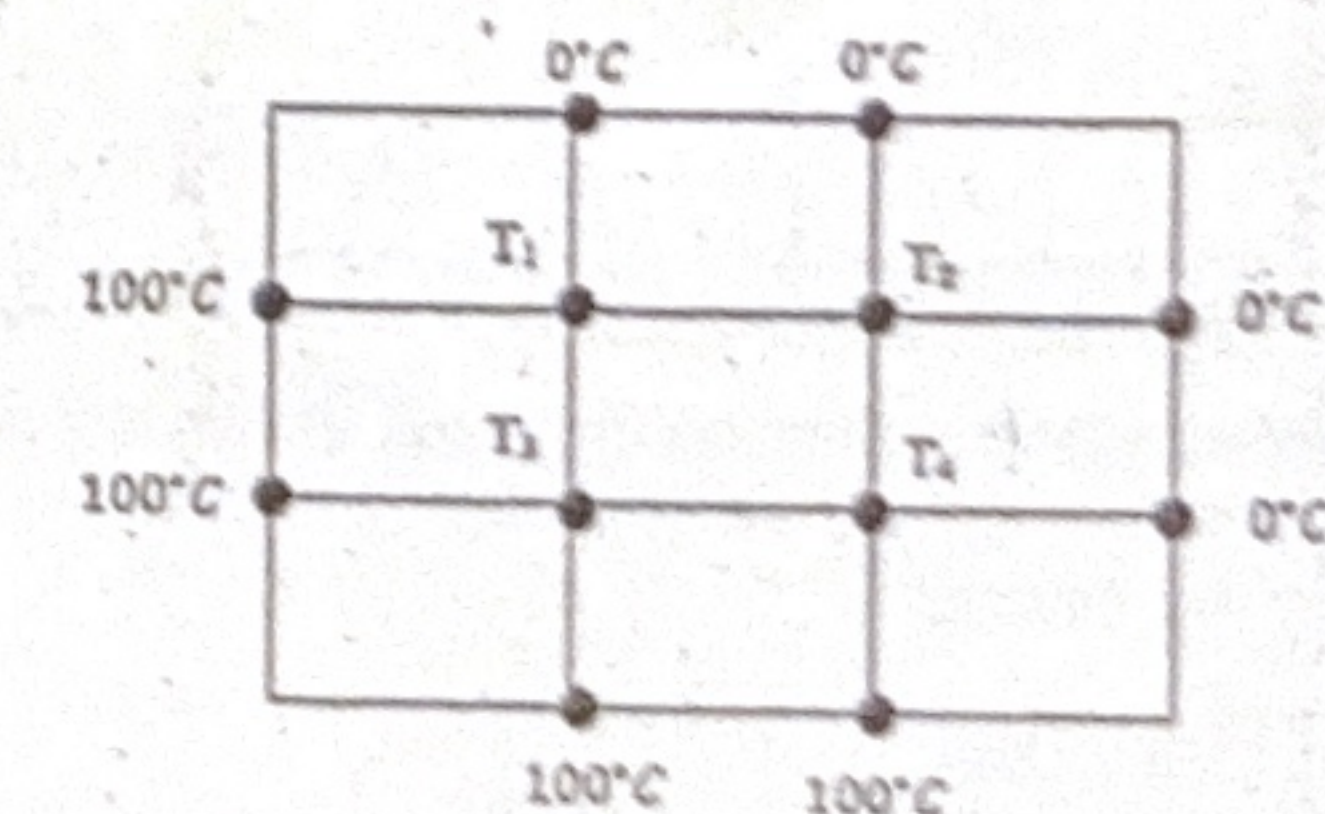
$$\begin{bmatrix} 3 & -1 & 0 \\ -2 & 4 & -3 \\ 0 & -1 & 1 \end{bmatrix}$$

5. a) Solve the following differential equation within $0 \leq x \leq 0.3$ using RK-4TH order Method. 8

$$10 \frac{dy}{dx} = x^2 + y^2, y(0) = 1 \text{ with } h = 0.1$$

- b) Apply Euler's method to approximate value of $y(0.3)$ for the differential equation: $\frac{dy}{dx} = y + x, y(0) = 1$. 7

6. a) For square bar of size $15\text{cm} \times 15\text{cm}$, calculate the steady state temperature at interior point for the grid size of $5\text{cm} \times 5\text{cm}$. 8



- b) Solve the Poisson equation $\nabla^2 f = 2x^2 + y$, over the square domain $1 \leq x \leq 4, 1 \leq y \leq 4$, with $f=0$ on the boundary. Take step size in x and $y, h=k=1$. 7

7. Write short notes on: (Any two) 2x5

- a) Errors in Numerical Method
- b) Ill-conditioned systems
- c) Boundary value problem

POKHARA UNIVERSITY

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Semester: Fall

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Attempt all the questions.

1. a) Write down the drawbacks of NR-method. Find a real root of the equation: $x^2 - 3x + 2 = 0$ by using NR- method in the neighbourhood of $x = 0$ correct to five significant digits. 8
- b) What is the difference between the convergence and divergence of a non-linear equation? Prove the oscillatory divergence and monotone divergence of the function: $f(x) = x^2 - 18$ by using Fixed point iterative method. 7
2. a) State an algorithm and convergence of second method. 7
- b) Estimate the value of $\sin \theta$ at $\theta = 25$ using newtons forward interpolation. 8

θ	10	20	30	40	50
$\sin \theta$	0.1736	0.3420	0.5	0.6428	0.7660

3. a) The growth of bacteria(N) in a culture after t hours is given by the following table 7

Time t(hr)	0	1	2	3	4
Bacteria(N)	32	47	65	92	132

If the relationship between bacteria N and time t is of the form $N = ab^t$. Using least square approximation estimate the N at $t = 5$ hr.

- b) Integrate the given integral $\int_0^{\frac{\pi}{2}} \frac{\cos x}{\sqrt{1+\sin x}} dx$ using Gauss Quadrature Formula for $N=2$ and $N=3$. 8
4. a) Find the inverse of matrix using Gauss Jordan Method. 7

$$\begin{bmatrix} 1 & 1 & 3 \\ 3 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$$

- b) Find the largest Eigen value and corresponding Eigen vector of the matrix. 8

$$A = \begin{bmatrix} 1 & 4 & 4 \\ 4 & 1 & 8 \\ 4 & 8 & 1 \end{bmatrix} \text{ using Power Method.}$$

5. a) Solve the following system of equations by using relaxation method correct to two decimal places. 8

$$9x - y + 2z = 9$$

$$x + 10y - 2z = 15$$

$$2x - 2y - 13z = -17$$

- b) Use Taylor series method to solve the ordinary differential equation: $dy/dx = x^2 + y^2$ for $x = 0.5$ subject to the initial condition: $y(0) = 1$. 7
6. a) Solve: $\frac{dy}{dx} = x + z$, $\frac{dz}{dx} = x - y$ for $x = 1.5$, given that $y = z = 1$ at $x = 1$ 8
- b) Solve Poisson's equation $u_{xx} + u_{yy} = 243(x^2 + y^2)$ over a square domain $0 \leq x \leq 1$, $0 \leq y \leq 1$ with step size $h = 1/3$ with $u = 100$ on the boundary. 7

7. Write short notes on: (Any two) 2.5

- a) Newton's differentiation formulas
- b) Overview of PDEs
- c) Shooting method

POKHARA UNIVERSITY

Level: Bachelor

Programme: BE

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Semester: Fall

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Full Marks : 100

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Attempt all the questions.

1. a) Explain the importance of Numerical Methods in Engineering Fields. Describe the different types of Error in Numerical Methods with an example. 7
- b) Find the root of equation $\cos x = xe^x$ using bisection method correct to four decimal places. 8

OR

Find the root of equation $2x - \log x = 7$ using Secant method correct to three decimal places.

2. a) Fit a power function model of the form $y=ax^b$ from the following data. 8

x	1	2	3	4	5
y	0.5	2	4.5	8	12.5

- b) Use lagrange's interpolation to find the value of y when x=3 from the following table. 7

x	3.2	2.7	1.0	4.8	5.6
y	22	17.8	14.2	38.3	51.7

OR

Estimate the value of $\sin \theta$ at $\theta = 25$ using newtons forward interpolation.

θ	10	20	30	40	50
$\sin \theta$	0.1736	0.3420	0.5	0.6428	0.7660

3. a) Use the Simpson's 1/3 rule to estimate the integral $\int_0^2 e^{x^2} dx$ taking 8 number of intervals. 8
- b) Use Gauss-Legendre 3-point formula to evaluate $\int_{0.5}^{1.5} e^{x^2} dx$. 7

4. a) Use power method to find the smallest Eigen value and Eigen vector of the following matrix: 8

$$A = \begin{bmatrix} 15 & -4 & -3 \\ -10 & 12 & -6 \\ -20 & 4 & -2 \end{bmatrix}$$

- b) Solve the system of equations by Gauss Jacobi Iteration method: 7

$$20x + y - 2z = 17$$

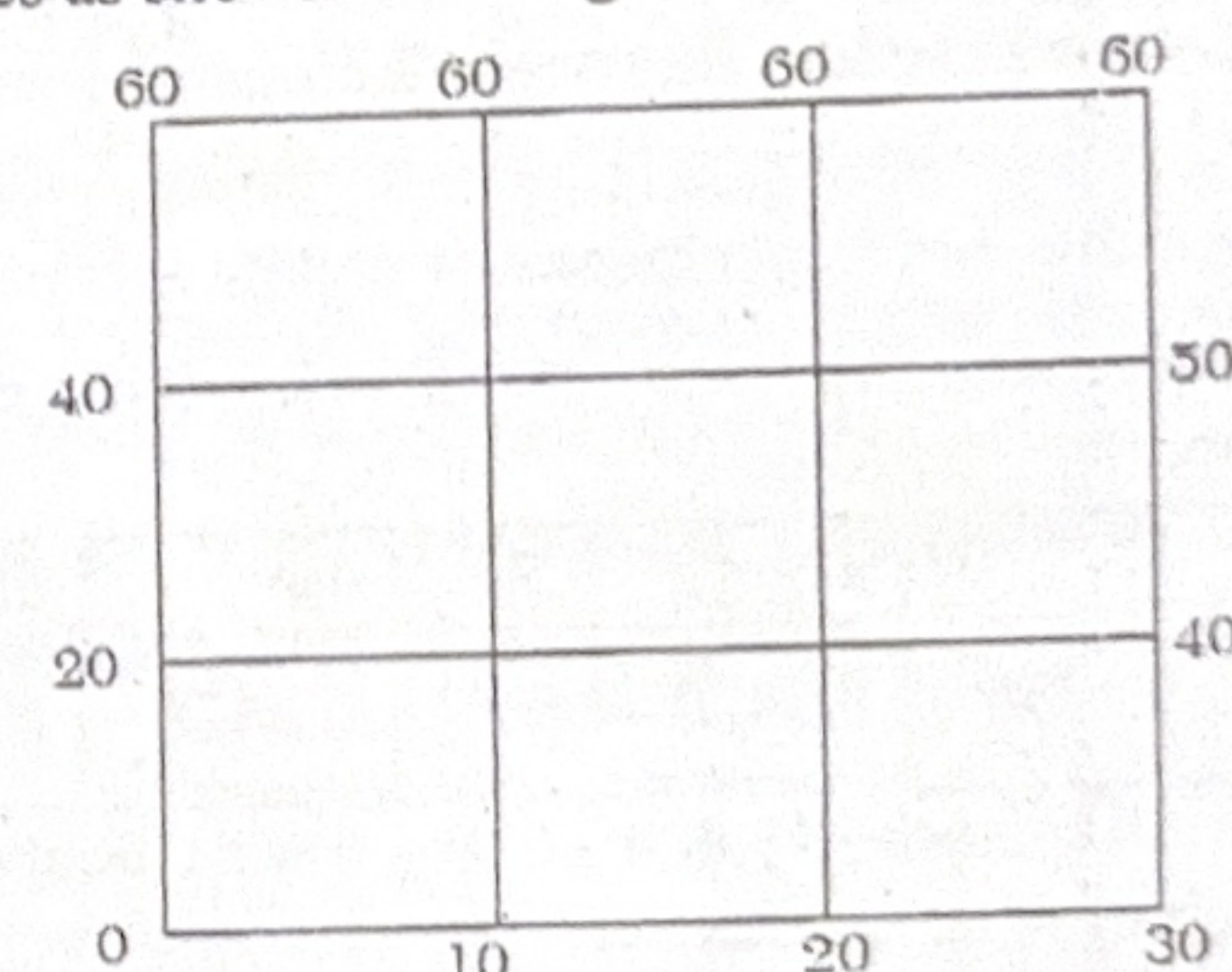
$$2x - 3y + 20z = 25$$

$$3x + 20y - z = -18$$

5. a) Solve $dy/dx = x+y$, given $y(0)=1$, when $x=0.3$ taking $h = 0.1$ using modified Eulers method. 7

- b) Applying runge kutta fourth order method, find an approximate value of y when $x=0.2$ given that: $y' = x + y$ and $y(0) = 1$. 8

6. a) Solve the equation $u_{xx} + u_{yy} = 0$ for the square mesh with the boundary values as shown in the figure. 7



- b) Solve the equation $\nabla^2 f = x^2 y + 2$ over the square domain $0 \leq x \leq 3, 0 \leq y \leq 3$, with $f = 0$ on the boundary and $h=1$. 8

7. Write short notes on: (Any two) 2x5

- a) Fixed point iteration method
- b) Fixed conditioned system of linear equation
- c) Shooting method

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Attempt all the questions.

1. a) What is the mathematical assumption being made for finding a real root of a non-linear equation? Find a real root of the equation: $\cos(x) + e^x + x^2 = 3$ by using Bisection method. 8

- b) What is the convergence of a non-linear equation? Find a real root of the equation: $e^{\cos x} - \sin(x) = 1$ by using NR-method such that the tolerance value is at least 0.000123. 7

2. a) The following table gives the population of a town during the last six censuses. Estimate the increase in the population during the period from 1965 to 1975. 8

Year:	1941	1951	1961	1971	1981	1991
Population:(in thousand)	12	15	20	27	39	52

- b) Fit the equation from the given set of data. 7

$$y = \frac{1}{ax + b}$$

x	-5	-4	-3	0	2	3	6	8
y	0.15	0.18	0.23	2	-0.5	-0.31	-0.143	-0.1

3. a) Apply Romberg's method to evaluate: $I = \int_0^2 \left(\frac{dx}{4+x^2} \right) dx$. 8

- b) Solve the following set of equation by Gauss Seidal method. 7

$$27x + 6y - z = 85$$

$$x + y + 54z = 110$$

$$6x + 15y + 2z = 72$$

4. a) Find the largest eigen value and corresponding eigen vector of given matrix using power method. 8

$$\begin{bmatrix} 1 & 4 & 4 \\ 4 & 1 & 8 \\ 4 & 8 & 1 \end{bmatrix}$$

- b) Solve the following system of linear equations using Gauss Elimination method (GEM) with partial pivoting technique. 7

$$y + 3u + 2v = 19$$

$$3y + 2u + 2v = 20$$

$$x + 4y + 2v = 17$$

$$-2x + 2y + u + v = 9$$

5. a) Solve the differential equation $y' = y - \frac{2x}{y}$ using Modified Eulers method within $0 \leq x \leq 0.2$ with initial condition $y(0) = 1$ and step size $h = 0.2$. 7

- b) Solve $y' = x + y$, $y(0) = 1$ by Taylor's series method. Hence find the value of y at $x = 0.1$ and $x = 0.2$. 8

6. a) Solve the equation: $\frac{d^2x}{dx^2} = 6xy^2 + y$, $y(0) = 1$ and $y'(0) = 0$ to find $y(0.2)$ and $y'(0.2)$, using RK-4 method. (Take $h = 0.2$) 7

- b) Solve the Poisson equation $\nabla^2 f = 2x^2 + y$, over the square domain $1 \leq x \leq 4$, $1 \leq y \leq 4$, with $f=0$ on the boundary. Take step size in x and y , $h=k=1$. 8

7. Write short notes on: (Any two) 2×5

- a) Fixed point iteration method

- b) Newton's differentiation formula

- c) Overview of PDEs

POKHARA UNIVERSITY

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Attempt all the questions.

1. a) Find the root of the equation $x^3 + x^2 - 3x - 3 = 0$ correct up to three decimal place using Bisection method. 7
- b) What are the limitations of NR method? Using NR method, find the root of $\log x - \cos x = 0$ correct up to three decimal places. 8

OR

Write an algorithm to find a real root of a non-linear equation using secant method.

2. a) The following table gives the population of a town during the last six censuses. Estimate the increase in the population during the period from 1976. 8

Year:	1941	1951	1961	1971	1981	1991
Population:(in thousand)	12	15	20	27	39	52

OR

The velocity of a rocket is measured at three different times as follows:

Time, t(s)	Velocity, v(t) (m/s)
10	227.04
15	362.78
20	517.35

Using a quadratic Lagrange interpolating polynomial, determine time t at which the rocket's velocity is 300m/s.

- b) Fit the equation from the given set of data.

$$y = \frac{1}{ax + b}$$

x	-5	-4	-3	0	2	3	6	8
y	0.15	0.18	0.23	2	-0.5	-0.31	-0.143	-0.1

3. a) Evaluate the integral $I = \int_4^{5.2} \log x \, dx$ for n = 6 using for simpsons 1/3 rule and 3/8 rule. Compare the result in both conditions. 7
- b) Evaluate the integral $\int_{0.5}^{1.5} e^{x^2} \, dx$ using the 3 point Gaussian quadrature. 8
4. a) Using the LU Crout decomposition method solve the following system of equations. 8

$$\begin{aligned} 4x_1 - x_2 + x_3 &= 7 & -2x_1 + 6x_2 + x_3 &= 9 & -x_1 + x_2 + 7x_3 &= -6 \end{aligned}$$
- b) Find the solution of the given simultaneous linear equation using Gauss Seidel method. 7

$$\begin{aligned} 6x_1 - 2x_2 + x_3 &= 11 \\ -2x_1 + 7x_2 + 2x_3 &= 5 \\ x_1 + 2x_2 - 5x_3 &= -1 \end{aligned}$$
5. a) Find the solution of the given ordinary differential equation at $x=0.5$ using the step size of $h=0.25$ using RK4 method. 7

$$\frac{dy}{dx} = x + y; y(0) = 1$$
- b) Solve the following equation for $y(0.2)$ using shooting method: 8

$$\frac{d^2y}{dx^2} + 2 \frac{dy}{dx} - 3y = 6x \text{ Given } y(0) = 0, y'(0) = 1.$$
6. a) Solve the Poisson equation $\nabla^2 f = 2x^2y^2$ over the square domain $0 \leq x \leq 3, 0 \leq y \leq 3$, with $f = 0$ on the boundary and $h = 1$. 7
- b) Solve the elliptic equation $U_{xx} + U_{yy} = 0$ over a square mesh of side four units satisfying the following boundary conditions; 8

$$\begin{aligned} u(0, y) &= 0 \text{ for } 0 \leq y \leq 4, & u(4, y) &= 12 + y \text{ for } 0 \leq y \leq 4, \\ u(x, 0) &= 3x \text{ for } 0 \leq x \leq 4, & u(x, 4) &= x^2 \text{ for } 0 \leq x \leq 4. \end{aligned}$$
7. Write short notes on: (Any two) 2x5
 - a) Importance of Numerical methods
 - b) Power Method
 - c) Review of ODEs

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Using bisection method, find where the graph $y = x - 6$ and $y = \ln(x)$ intersects. Write your answer correct to three decimal places. 7

OR

Find the real root of $x^5 - 3x^3 - 1 = 0$ correct to four decimal places using secant method.

- b) Find the square root of 7 using Newton Raphson method correct to 4 decimal places. 8
2. a) Estimate the value of $\sin\theta$ at $\theta = 45$ using Newtons backward interpolation. 7

θ :	10	20	30	40	50
$\sin\theta$:	0.1736	0.3420	0.5	0.6428	0.7660

OR

Fit a second order polynomial to the following data.

x :	1	2	3	4
y :	6	11	18	27

- b) Find a cubic polynomial $f(x)$ from the following table using Lagrange interpolation method. Also, use the polynomial to find $f(1.5)$. 8

x :	0	1	2	3
$f(x)$:	1	2	1	10

3. a) The distance covered by a particle at any time is given in the table below. 7

Time (t, sec):	5	6	7	8
Distance (d, m):	10.0	18.6	23.5	33.4

Calculate the velocity of the particle at $t = 6$ sec.

- b) Evaluate the integral $I = \int_0^{0.6} e^{x^2} dx$ for $n = 6$, by using 8

- Trapezoidal rule.
- Simpsons $1/3^{\text{rd}}$ rule.
- Simpsons $3/8^{\text{th}}$ rule.

4. a) Solve the following system of equations using Doolittle's method. 7

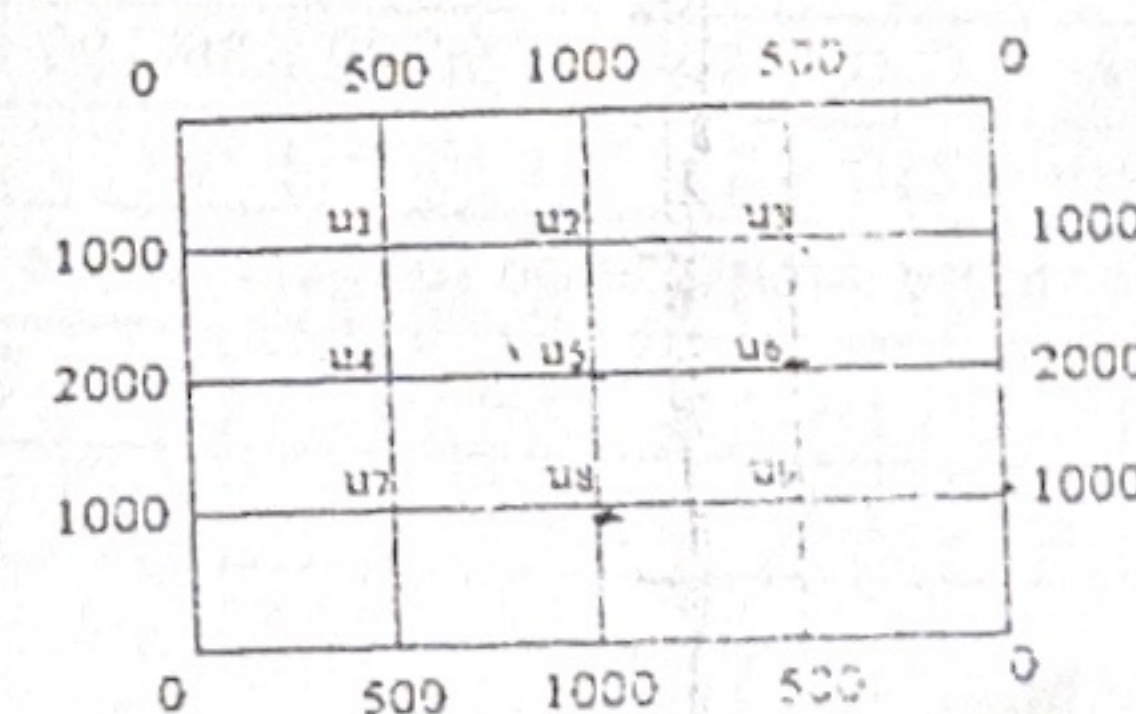
$$\begin{aligned} 2x + 3y + z &= 9, \\ x + 2y + 3z &= 6, \\ 3x + y + 2z &= 8. \end{aligned}$$

- b) Find the largest Eigen value and corresponding Eigen vector from the following matrix $\begin{pmatrix} 2 & 4 & 1 \\ 0 & 2 & 3 \\ 1 & 0 & 3 \end{pmatrix}$ using power method. 8

5. a) Solve by using Euler's method $\frac{dy}{dx} = \frac{2x + e^x}{x^2 + xe^x}$ given $y(1) = 0$, for y at $x = 1.04$ taking $h = 0.01$. 7

- b) Solve the differential equation by RK-4 method. 8
- $y'' - xy' + y = 0$ with initial condition $y(0) = 3, y'(0) = 0$ for $y(0.2)$ taking $h = 0.2$.

6. a) Solve the elliptic equation $u_{xx} + u_{yy} = 0$ for the following square mesh with boundary values as shown in figure. 7



- b) Solve Poisson's equation $u_{xx} + u_{yy} = (2 + x^2y)$ over a square domain of $0 \leq x \leq 3$ and $0 \leq y \leq 3$ with $u = 0$ on the boundary and mesh length $h = 1$. 8

7. Write short notes on: (Any two) 2x5

- Importance of numerical computation in the field of science and engineering.
- Ill conditioned and well conditioned system
- Application of interpolation

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Numerical Methods

Semester: Fall

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Find the root of the equation $f(x) = x^2 - 4x - 10$ correct to three decimal places by using False Position method. 7
- b) From the data given below, find the number of students whose weight is between 60 to 70. 8

Weight in lbs	0-40	40-60	60-80	80-100	100-120
No. of students	250	120	100	70	50

2. a) Find the cubic spline interpolation formula for the following data: 8

x	1	2	3	4	5
f(x)	1	0	1	0	1

- b) Evaluate the integral $I = \int_0^{\frac{\pi}{2}} (1 + 3 \cos 2x) dx$. Compare the result in both conditions for Simpson 1/3 and 3/8 rule. 7

3. a) Use Romberg method to evaluate: 8

$$\int_0^{3\pi/2} e^x \sin x \, dx$$

- b) Solve the following system of equations using partial pivoting method. 7

$$x + y + z = 4, \quad x + 4y + 3z = 8, \quad x + 6y + 2z = 6$$

4. a) Find the solution of the given simultaneous linear equation using Gauss Seidel method. 8

$$\begin{aligned} 6x_1 - 2x_2 + x_3 &= 11 \\ -2x_1 + 7x_2 + 2x_3 &= 5 \\ x_1 + 2x_2 - 5x_3 &= -1 \end{aligned}$$

- b) Using Dollittle LU decomposition method, solve the following system equations: 7

$$3x + 2y + z = 10$$

$$2x + 3y + 2z = 14$$

$$x + 2y + 3z = 14$$

5. a) Using the R-K 1st order method, find an approximate value of y corresponding to $x = 1$, given that $\frac{dy}{dx} = \frac{y-x}{y+x}$ and $y = 1$. When $x = 0$, and $h = 0.02$. 8

- b) Find the solution of the given ordinary differential equation at $x = 0.5$ using the step size of $h=0.25$ using Heun's method. 8

$$\frac{dy}{dx} + 0.4y = 3e^{-x}, \quad y(0) = 5$$

6. a) Solve the given differential equation by RK-4th order method $y'' - x^2 y' - 2xy = 0$ with initial condition $y(0) = 1, y'(0) = 0$. for $y(0.1)$ taking $h = 0.1$. 8

- b) Solve the Poisson equation $\nabla^2 f = 2x^2 + y$, over the square domain $1 \leq x \leq 4, 1 \leq y \leq 4$, with $f=0$ on the boundary. Take step size in x and $y, h = k = 1$. 7

7. Write short notes on: (Any two)

- a) Error in Numerical method
- b) Algorithm of fixed point iteration method
- c) Classification of Second Order Partial Differential Equation